

The Data Dojo

A Power BI

Community of Practice

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DATA POINT

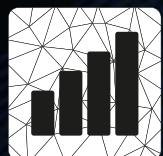
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Partners

Data Point Prague 2024



Data
Brothers



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About the Speaker

Intro

- Name: James D. Bartlett III ([@JamesDBartlett3](#)) {he/him/his}
- Title: Senior BI Analyst at [Des Moines University](#) in Iowa, USA

Professional Credentials

- [Microsoft MVP \(Most Valuable Professional\)](#)
- Microsoft Certifications
 - [Microsoft Certified: Fabric Analytics Engineer Associate \(DP-600\)](#)
 - [Microsoft Certified: Power BI Data Analyst Associate \(PL-300\)](#)

Extracurricular Activities

- User Groups
 - [Iowa Power BI User Group \(Co-Organizer\)](#)
 - [Higher Education Power BI User Group \(Co-Organizer\)](#)
- Open-Source Projects
 - [Fabric-Archive-Bot](#)
 - [ps-for-pbi](#)
 - [PowerBits](#)
 - [PoshBits](#)

Opinions expressed are solely my own, and not the views of my employer, nor of any other organization with which I am affiliated.



Agenda

Founding the Data Dojo

Calling All Data Ninjas

Workshops 1 & 2

Our First Major Setback

The Data Dojo Strikes Back!

Workshops 3 – 5

Office Hours & Coffee Lounge

Resources

Q&A



Founding the Data Dojo



What is a Community of Practice?

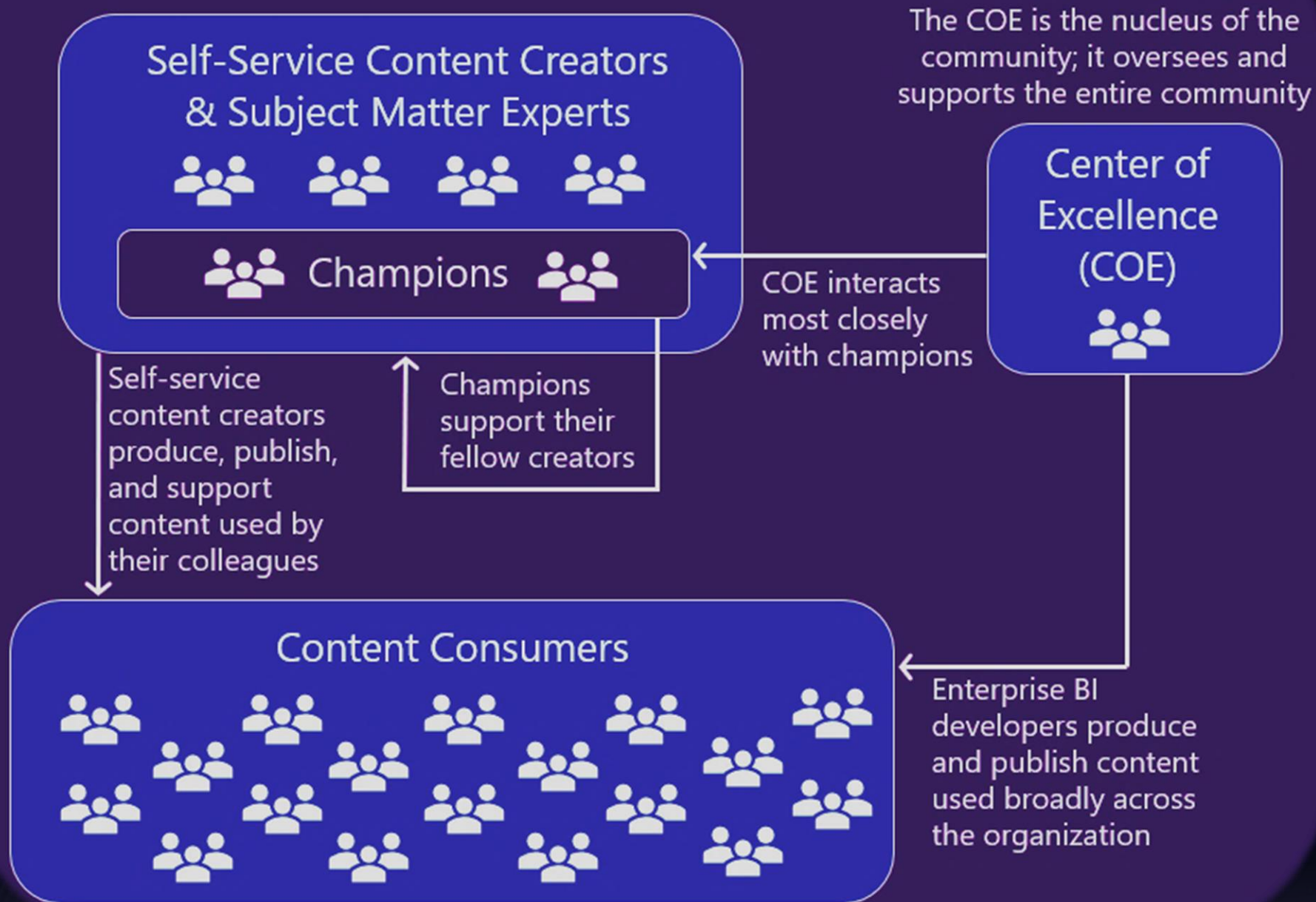
“Communities of Practice are groups of people who share a concern or a passion for something they do and learn how to do it better as they interact regularly.”

– Etienne and Beverly Wenger-Trayner, June 2015



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/* ----- Note ----- */  
'Community of Practice' <> 'Center of Excellence'
```

Power BI Community of Practice



Data Dojo Backstory Part 1: The Status Quo

- Des Moines University (DMU) is a graduate-level medical school
- Most faculty & staff have advanced degrees in STEM
- Strong culture of evidence-based, data-driven decision making
- Not-for-profit org, so budget is often a limiting factor
- Huge backlog of data projects waiting for time and/or funding

Data Dojo Backstory Part 2: A New Paradigm

- Decentralized, democratized data analytics development
- Inter-disciplinary knowledge sharing
- Intra-organizational communication
- Cross-functional, silo-busting collaboration

Founding Principles of the Data Dojo

1. We welcome anyone to join – no gatekeeping allowed!
2. Our events & activities are 100% optional
3. Our members are encouraged (but never required) to share knowledge and collaborate
4. We facilitate collaboration & knowledge sharing between our members via:
 - Interactive events (in-person, virtual, and hybrid)
 - A private Viva Engage group

Founding Principles of the Data Dojo

- 5. We have no instructors or students – we all learn from each other
- 6. We believe the only “stupid question” is the one that goes unasked
- 7. We vote on dates & topics for upcoming events
- 8. Our members are encouraged to network & build intra-organizational relationships

Calling All Data Ninjas



Recruiting Data Ninjas (Data Dojo Members)

1. Create a Microsoft Forms survey to gauge respondent's interest in the Data Dojo, availability, and current skill level
2. PBI Monitor report
 - List all users with active Power BI Pro licenses in the organization
 - Filter out users with no activity logs from previous 12 months
3. Send Viva Engage group invite and MS Forms survey link to top 20% of Power BI users by average monthly activity count

Workshops

1 & 2



Workshop #1

1. Introductions around the room
2. What we're doing here, and why it matters
3. Data growth – size, significance, sources, silos, and solutions
4. The importance of speed and accuracy in data analysis
(*and what we mean by “speed” and “accuracy”*)
5. Data driven decision makers:
 - What they do know
 - What they often don't know

Workshop #1

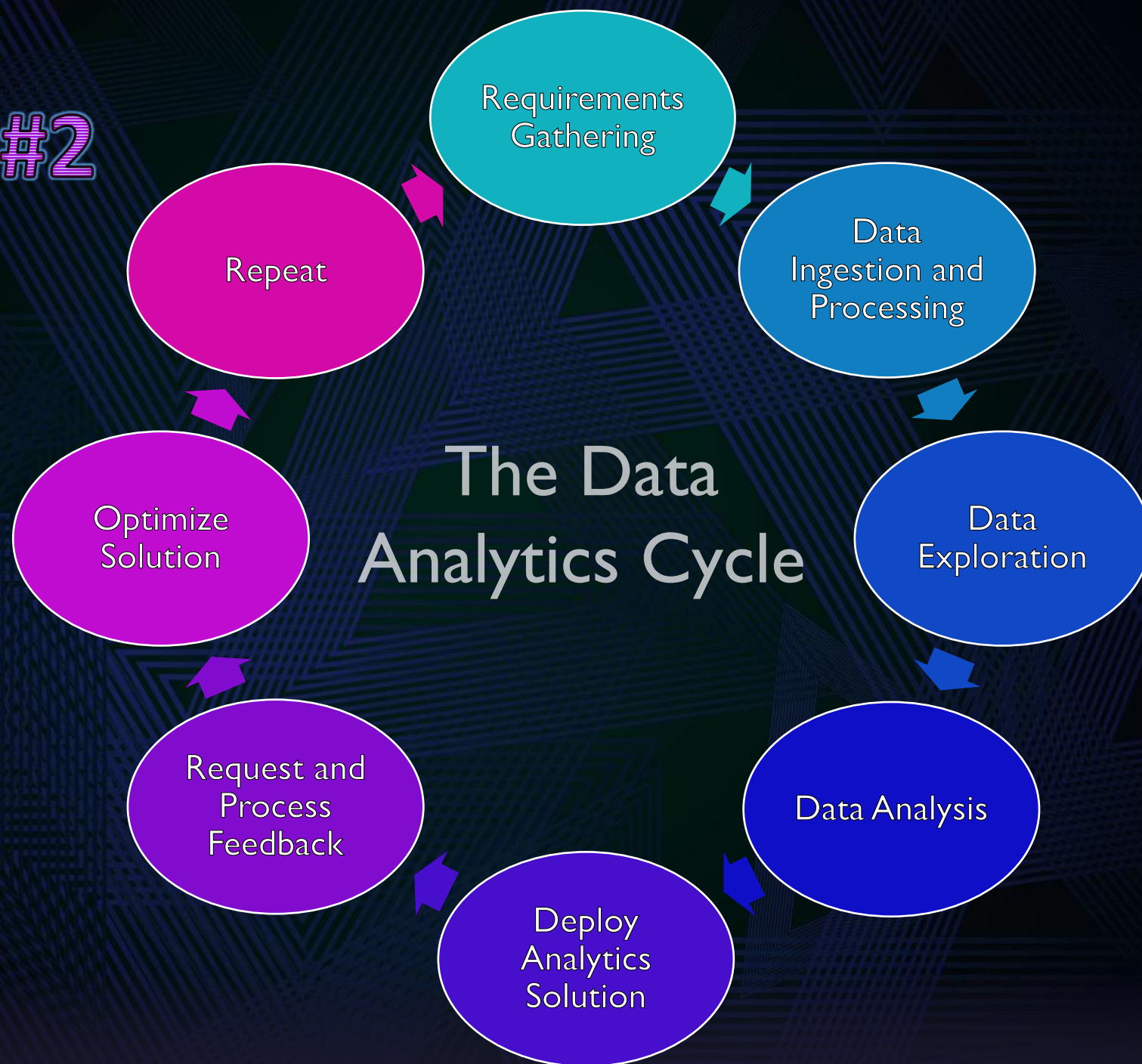
6. The current paradigm and state of data analysis at DMU
7. The new paradigm – our vision for the future, and plans for turning that vision into a reality
8. Testimonials from 4 Data Dojo members with prior Power BI experience

Workshop #2

The 5 Types of Data Analytics

1. Descriptive – “What happened, where, and when?”
2. Diagnostic – “Why did it happen?”
3. Predictive – “What will happen in the future?”
4. Prescriptive – “What should we do about it?”
5. Cognitive – “How can we do it better?”

Workshop #2



Workshop #2

- Structured vs. Unstructured Data
- Quantitative vs. Qualitative Data
- “Better Instruction Through Data” (member presentation)
 1. Live demo of reports made using curated semantic models provided by IT department
 2. Stories of how the insights from these reports are making an impact in the business
 3. Challenges encountered and overcome through knowledge sharing and collaboration

Our First Major Setback



Pitfalls on the Path

My 3 Big Mistakes

1. Inconsistent Schedule
2. Inconsistent Engagement
3. Letting “Perfect” be the Enemy of “Good”

Other Contributing Factors

- New Campus Projects
- Loss of Project Manager

Mistake #1: Inconsistent Schedule

1. Plenty of momentum, but no roadmap
2. Tried to find dates that would work for all members
3. Kept delaying next event indefinitely
4. Failed to communicate with members about these delays

Mistake #2: Inconsistent Engagement

1. Failed to keep recruiting new members
2. Failed to consistently engage with existing members
 - Didn't elicit feedback and suggestions for future topics from members
3. Failed to seed community discussion with new material
 - Viva Engage group activity mostly between same handful of members

Mistake #3: Letting Perfect be the Enemy of Good

1. Had a “grand vision” for the next event

- Introduce the Power BI Version Control app
- Deploy latest version to all interested org units

2. Failed to anticipate delays in release schedule

3. Again, kept delaying next event indefinitely

4. Perfectionism obstructed progress

“No plan of operations can with any certainty reach beyond the first encounter with the enemy.”

- Helmuth von Moltke, 1871

The Data Dojo Strikes Back!



The Art of War: Strategy First, Then Tactics

- Zoom out, re-calibrate, and then zoom back in
- Target attainable goals first, then set “stretch goals”
- Beware of “tunnel vision” masquerading as “laser focus”

“Strategy without tactics is the slowest route to victory.
Tactics without strategy is the noise before defeat.”

– Sun Tzu, *“The Art of War”*



Wax On, Wax Off

- Master the Fundamentals
 - Develop and refine leadership skills
 - Research and implement CoP best practices
- Develop Core Strength
 - Skill up at every opportunity
 - Share knowledge with anyone who expresses interest
- Practice Related Disciplines
 - Event planning
 - Project management



Wake the Sleeping Dragon

- Begin scheduling events ASAP
- Engage or communicate with the Data Dojo at least 2x per week
- Recruit 2 “Data Dojo Deputies” from existing members
- Encourage members to “show and tell” their projects



Workshops

3 - 5



Workshop #3: Technologies & Terminologies

- Data Governance, Data Stewards, Data Lineage
- Data Sources (ODS, OLTP, OLAP, CSV, API, etc.)
- On-Premises vs. Cloud
- ETL (Rules, Relationships, Routines)
- Fact Tables, Dimension Tables, Views, Stored Procedures
- Data Warehouse/Lakehouse, Dataflow, Semantic Model, Report
- Star Schema, Snowflake Schema, Spaghetti Schema
- Calculated Tables, Columns, and Values (Measures)

Workshop #3: Technologies & Terminologies

- Report vs. Dashboard
- Static/Paginated Reports vs. Interactive/Canvas Reports
- KPIs & OKRs
- Alerts & Subscriptions
- Workspaces, Roles, Apps, Security Groups, Audiences
- Row-Level Security (RLS) vs. Object-Level Security (OLS)
- Version Control (SharePoint, OneDrive, Git Integration, etc.)

Workshop #4: Power BI Desktop Overview

- Installing Power BI Desktop from the Microsoft Store
- Introducing and Touring the Data Dojo Report Template
 - Dummy financial data (flat file, included with Power BI Desktop)
 - Normalized in Power Query
 - Modeled as a star schema in Model View
 - Measures
 - Implicit vs. Explicit
 - How to write a simple explicit measure in DAX
 - Filters
 - Filter Pane Sections: Report-Level, Page-Level, and Visual-Level
 - Report-Level (Filter Pane) vs. Model-Level (Power Query)

Workshop #4: Power BI Desktop Overview

- Touring the Data Dojo Report Template (*continued*)
 - Slicers
 - Layout: Vertical vs. Horizontal
 - Style: Single-select, multi-select, range, etc.
 - Built-in vs. custom
 - Standard on-canvas placement vs. bookmark-driven show/hide pane
 - Slicers vs. Filter Pane
 - Relative strengths and weaknesses
 - When and where to use each kind
 - Sorting: Ascending, Descending, and “Order by Column”

Workshop #5: Import and Transform Data

Data Dojo Report Template – Advanced Features

- Data Freshness Indicator (DFI)
- Utility Tables
 - __DataSource_Info
 - __SemanticModel_Info
- Power Query (M) Parameters
 - Source_Workspace
 - Source_Dataflow
 - Row_Limit
- Custom Power Query (M) functions
 - fn_GetTableFromDataflow
 - fn_GetLocalTime
 - fn_Coalesce

The Importance of Hands-On Workshops

1. Slideshows are *boring*

- Boring to make
- Boring to watch – *except **this** one, of course!* 😄

2. Technical concepts and skills are best learned...

- Via live demonstration
- In an interactive, hands-on format

3. Following along with a live demo is...

- More memorable (and thus more effective as a learning aid)
- More fun!

Office Hours & Coffee Lounge

A New Data Dojo
Event Format



Traditional Power BI Workshops

- Hosting a traditional Power BI workshop requires lots of...
 - Time
 - Energy
 - Patience
 - Coordination
- None of these resources are in plentiful supply
- Maintaining a consistent schedule of workshops is difficult at best
- But Data Dojo members need to keep learning and stay engaged!
- So, what's a Data Dojo organizer to do?

The Data Dojo: Office Hours & Coffee Lounge

Inspired by 2 scenes often found on a university campus

1. Office Hours

- Consistent schedule of days and times each week
- Members can visit at any time during those hours
- No specific topics planned, just answering any questions the members may have
- If no one shows up, the organizer can work on their usual tasks

2. Coffee Lounge

- A place to gather, hang out, and have discussions
- A casual environment with no expectations
- Socializing is encouraged (it helps build and strengthen relationships)

Resources



Resources for Data Dojo Organizers

- Community of Practice

- wenger-trayner.com/introduction-to-communities-of-practice
- learn.microsoft.com/power-bi/guidance/powerbi-adoption-roadmap-community-of-practice

- Power BI Adoption Roadmap

- learn.microsoft.com/power-bi/guidance/powerbi-adoption-roadmap-overview

- DataVolume Blog

- datavolume.xyz/tags/#Community+of+Practice

- PBI Monitor: Power BI activity log analytics

- github.com/RuiRomano/pbimonitor



Resources for Data Dojo Members

- Power BI Learning Resources

- [datavolume.xyz/assets/doc/DMU Data Dojo - Power BI Learning Resources.pdf](#)
- [datavolume.xyz/assets/doc/DMU Data Dojo - Power BI Learning Resources.xlsx](#)

- Slide Decks from Workshops 1 & 2

- [datavolume.xyz/assets/doc/DMU Data Dojo - Meeting 001.pdf](#)
- [datavolume.xyz/assets/doc/DMU Data Dojo - Meeting 002.pdf](#)



Q & A





Thank You

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